PIER

PUEY UNGPHAKORN INSTITUTE
FOR ECONOMIC RESEARCH

Glancing at Labour Market Mismatch with User-generated Internet Data

by

Nuarpear Warn Lekfuangfu, Voraprapa Nakavachara
and Paphatsorn Sawaengsuksant

January 2017

Discussion Paper

No. 53

The opinions expressed in this discussion paper are those of the author(s) and should not be attributed to the Puey Ungphakorn Institute for Economic Research.

Two sides of the same coin:

Glancing at labour market mismatch with user-generated internet data

Nuarpear Warn Lekfuangfu[1], Voraprapa Nakavachara and Paphatsorn Sawaengsuksant

Working Paper: 15 December 2016

Abstract

In this project, we will conduct a series of research exercise to demonstrate how selected web-based data sources can provide additional insights for labour market analysis, beyond what conventional government-conducted surveys can offer. We exploit web-based data from selected job-boards and resume postings under Thai domain to provide some insights on job vacancy statistics, labour market mismatch between required skill *vis-a-vis* attained skill at occupation level and the gap between reservation wage and productivity. We also test for potential impacts of the 300-baht minimum wage increase in 2013 and find negative relationship with our measure of province-level labour market tightness. We also use this dataset to investigate labour market discriminations using separate perspective of firms and job seekers.

Keywords: Internet job search, Employment outcomes, Mismatch**,** Minimum Wage

JEL Codes: J2, J64

[1] Lekfuangfu and Nakavachara are lecturers at Chulalongkorn University, Faculty of Economics. Lekfuangfu is a research associate at the Centre for Economic Performance (LSE). Sawaengsuksant is at the Bank of Thailand. We are grateful to insightful comments offered by Piti Disyatat, Nasha Ananchotikul, Attakrit Leckcivilize, Piriya Pholphirul and seminar participants at BOT and Chulalongkorn. The authors would like to thank Donn Tawanpitak and Pakawin Mansukmanee for their excellent assistance with the data. The results and findings reflect the personal view of the authors and do not reflect the views of their professional institutions.

I. Background of conventional labour market indicators of Thailand

The advancement of the Internet, along with its growing influence on every day's life have made possible for researchers to use it as a platform for potential data sources stretching across multiple domains. Research in economics have already benefited from new data sources and online research platform, ranging from Google Trend, Amazon Mechanical Turk and social network mobile information (Choi and Varian 2009, Horton et al. 2011, Hooley et al, 2012). In their seminal papers, both Kuhn and Skuterud (2004) and Askitas and Zimmermann (2009, 2015) acknowledge the priceless contribution of the online world on research in labour economics. Furthermore, user-generated online data are available in real time and therefore allows researchers to investigate current trends and characteristics without time lag.

In Thailand, there are a number of sources that help provide meaningful insights to labour market conditions of the economy. To track labour market behaviours at the macro-level, policy makers conventionally rely on the statistics drawn from the monthly Labour Force Survey (LFS) to depict the states of employment equilibrium of the country- more famously the unemployment rate. To gain fuller comprehension of the demand and the supply side of the market separately, more statistics are drawn from various sources, which survey selected sectors in the economy.[2] In addition, published statistics from administrative data from the Ministry of Labour (MoL) provide the aggregate vacancies reported by firms as well as the monthly total dismissals nation-wide derived from the claimants of the social security jobless benefits.[3]

We recognise that the conventional sources of information, in particular the LFS, remains significantly valuable to provide insights to labour market conditions nationally. One notably advantage of this survey is that it has been conducted for a long period of time- therefore this consistent continuation allows researchers and policy makers to understand the dynamic of Thai labour markets many decades back. Moreover, as a nationally representative

[2] The Business Sentiment Index (BSI) survey privates companies and calculate diffusion index of current hiring situation and hiring plans in next 3 months. The Labour Demand of Establishment Surveys (LD) survey medium-to-large enterprises in 8 sub-sections of manufacturing and some services. For the latest survey in 2013, it covered establishments of 6 employees or more for selected sectors. This counts as 31,856 establishments nation-wide. The sectors covered in the survey are: commercial, service, construction, manufacturing, land transportation, tourism and hospitals.

[3] Currently, the MoL also produces its own labour market indicators, under the name of Labour Market Warning. This index includes the statistics mentioned above and also other industries' macro-level indicators. Very interestingly, it includes the total number of job advertisement. But this is also the number taken from print outlets. Given the downward trend of prints, this number is believed to hugely under-represent the actual total job ads at a point in time.

survey, one can constructively draw lesson learnt from statistical analysis using this dataset to the national platform.

However, it is worth recognising that a collection of statistics from the LFS, especially the employment status as well as salaries of workers, is the outcomes of the demand and supply interactions. That is, these are average salary or unemployment rates at the equilibrium of the market. While it tells a decent story of labour market in general, it cannot un-wrap the mechanisms within the markets. A number of advanced macro-economic models are generally embedded with a feature of labour market matching model (Mortensen and Pissarides, 1994; Shimer, 2005). This model requires information from each separate side of the same markets. While the LFS can provide the unemployment rate (u), it lacks information at all on the demand side – the vacancy rate (v). We note that one may use the LD or the MoL's vacancy report as the proxy. But, one must also note that these numbers are not only generated at completely different times, they are taken from different economic sectors and most of all, do not represent the same matched sampling pools.

In many other countries, there exists a number of ways in which the statistics for the crucial "labour market tightness (v/u)" can be calculated more properly[4]. For example, in France, researches make use of a matched employer-employee dataset- whereby the samplings are taken from the administrative data[5]. In the US, the Bureau of Labor Statistics produces, on top of the LFS equivalent (Current Population Surveys), the Job Openings Labor Turnover (JOLTS) as its complementary monthly survey for the vacancies from a sample of nonfarm establishments of entities in the 50 states[6]. JOLTS data are collected for total employment, job openings, hires, quits, layoffs and discharges, other separations, and total separations. However, it is worth notice that, as a survey, JOLTS sample under-

[4] Labour market tightness composes of two key features of the Beveridge Curve- vacancy and unemployed. When an economy is expanding, v generally gets bigger and u shrinks. Therefore the tightness (calculated as v/u) will get smaller, and vice versa. Historically, the US data shows a negative relationship between v and u. However, this relationship breaks down in the decent decade- where v may get bigger but not much change of u.

[5] For example, the D´eclaration Annuelle des Donn´ees Sociales (DADS) (DADS) is an administrative database of matched employer-employee information collected by the INSEE (Institut Nationale de la Statistique et des Etudes Economique). The data are based on the mandatory reports, filed by employers, of the gross earnings of each employee in compliance with French payroll taxes. All wage-paying individuals and legal entities established in France are required to file payroll declarations; only individuals employing civil servants are excluded from filing such declarations. But this dataset lacks vacancies.

[6] The JOLTS survey design is a stratified random sample of 16,000 nonfarm business and government establishments. The sample is stratified by ownership, region, industry sector, and establishment size class. The establishments are drawn from a universe of over 9.1 million establishments compiled by the Quarterly Census of Employment and Wages (QCEW) program which includes all employers subject to state unemployment insurance laws and federal agencies subject to the Unemployment Compensation for Federal Employees program

represents sectors or enterprises with high exit rate (fail within the first year) (BLS, 2016). This is mainly due to the time-consuming process of any surveys[7].

More recently, the US's Conference Board Help Wanted *Online* Data Series (HWOL) has been in the focus (Barnichon, 2010, 2015; Sahin, 2011). This is an updated series of the Conference Board Help Wanted Series, which report total number of job ads from print media. The attention directed to HWOL is a response to the recognition of the growth of digital activities in the US in the recent decades. Unlike the old version, HWOL consists of job ads from both the prints as well as from the online job boards across all US states. And unlike JOLTS, HWOL reports the actual total number of job vacancies of the entire pool of advertised job openings across number of industries, in a timelier manner[8].

Our research aims are to explain and understand behaviours of labour markets in Thailand from these alternative datasets. These chosen datasets in this research share three features. First, all datasets reflect the state of labour markets in Thailand in real-time. Second, they are virtually costless to collect and they are user-generated information. And third and foremost, these datasets reflects the growing shift towards digital activities and economies of markets in Thailand and elsewhere (Autor, 2001; Nakamura et al. 2009, Savankulov 2010).

As the world is becoming more digitized, more and more people rely on internet for their everyday activities. Currently there are 38 million internet users in Thailand which account for 56% of the total population (as of January 2016). The internet penetration rate for Thailand, defined as percentage of the population using internet divided by the total population, has increased significantly over the years, from 26% in 2014, 37% in 2015, to 56% in 2016. The penetration rate is currently 29% for Asia Pacific and 48% worldwide[9].

[7] Moreover, Jobs to be filled only by internal transfers, promotions, demotions, or recall from layoffs are excluded. Also excluded are jobs with start dates more than 30 days in the future. The separations level is the total number of employment terminations occurring at any time during the reference month, and is reported by type of separation—quits, layoffs and discharges, and other separations.

[8] The HWOL program contains data collected from over 16,000 online job sources including traditional job boards, corporate boards, and social media sites. Internet job sources that are aggregators (i.e. only scrape ads from other sources and provide no unique ads) are identified and removed from active collection in order to eliminate a major source of duplication in counting online ads. New job sources are identified using independent research and recommendations from industry sources across the U.S. This process results in periodic updates to the HWOL coverage. Job sources that cover smaller niche markets are also included in HWOL; however, smaller local job sources in an area with a limited number of ads may not be targeted for collection. The coverage are occupations to 2 digits, at regional and Metropolitan (MSA) level. (*https://www.conference-board.org/data/helpwantedonline.cfm)*

[9] Sources: Global Digital Statistics 2014, We Are Social SG; Digital Social & Mobile in 2015, We Are Social SG; Digital in 2016, We Are Social SG.

Based on the Information and Communication Technology Survey (ICT thereof) conducted by the National Statistical Office of Thailand (NSO) in 2015[10], there were just over 1 million people (aged 15 and older) reported using internet for job searches and application submissions. Most of them were among 20-29 years old (about 42%). Although the number of online job searchers may seem small compared to the total population, this number is not small compared to the number of unemployed people in quarter 1 of 2016 which was 370,000 (see Table 1.1). In addition, the fact that there were more people searching for jobs online compared to the number of people unemployed suggested that there could be quite a number of on-the-job searches.

Globally, the internet penetration rate has increased. Concurrently, there is also an increasing number of research papers that used internet-based data to explain various dimensions of the economy, and of our particular interest, to analyse the labour market. Lenaerts, Beblavy and Fabo (2016) review the literature on how various types of internet-based data such as Google Trends, LinkedIn, Facebook, Twitter, and Glassdoor Survey have been used or discussed in labour-related research. Antenucci, *et al* (2014) created the University of Michigan Social Media Job Loss Index based on job-related messages that people posted on their Twitter accounts. This index appeared to outperform the existing forecast for US initial claims for unemployment insurance. Marinescu (2016) utilizes the data from CareerBuilder.com, a large US internet job board, to analyse the impact of the increased unemployment benefits on job applications and job vacancies. For the Chinese labour market, Khun and Shen (2013) utilized the data from Zhaopin.com, the third largest internet job board in China, to analyse gender discrimination. More recently, Deming and Kahn (2016) exploit a near universe database of online job vacancies provided by Burning Glass Technologies and explain heterogeneity in wages across firms and labour markets. The same dataset has been used to identify types of skill shortages across sectors in the US (Burning Glass Technologies, 2015).

In additional, a collection of study look at the benefits of internet as a job search tool. For applicants, they find that job-seekers are more likely to get employed or have shorter

[10] Note that the internet penetration rate reported by this survey was 39.3% for 2015. In the previous paragraph we utilized the statistics by *We Are Social* since this source collected such information over time whereas the ICT survey only has the information for 2015. We acknowledged that the inconsistency between the two sources may be due to how they collected and processed their data.

unemployment duration if they also searched on internet.[11] The study by Kuhn and Skuterud (2004) are the exception. They use the US CPS Computer Supplements and find a relationship between shorter duration of unemployment and using internet as a tool. However, because of the effect disappeared after controlling for many observed individual characteristics, they eventually concluded that internet makes no difference. Fountain (2005) look at the same CPS but analysed the time periods separately. She finds that an early adopter of internet as a job search tool did benefit from using this mode but not the same is found for later cohorts.

Not only has the Internet revolution changed the behaviours of job seekers, it has also altered significantly the recruitment methods of many firms. A survey of the Fortune Global 500 shows that the year 2000, all of these top companies made use of e-recruiting services (Nakamura et al 2009). In comparison to print ads, online platforms have reduced the financial costs of filling vacancies and increased the reach to the public most significantly[12].

In the next section, we will explore the advantages of user-generated database from online-platforms of labour markets in understanding real activities in labour markets from the supply side (applicants) as well as the demand side (job ads) of the markets. Section 3 analyse relationships of different labour market conditions using the job portal datasets. Section 4 will investigate the nature of gender-targeting job ads and job applications in Thai labour markets.

II. Exploring datasets from internet-based job portals: sources and applicability

The existing survey datasets of labour markets in Thailand, in particular the LFS, is very valuable but yet remain limited for a number of reasons. First, they take time to prepare, collect and report. Therefore, there is an element of time lag in the dataset. Second, the respondents are propped to reply to the set of questionnaire, and most of the time in person. This leaves a lot of room for under-reporting non-favourable behaviours and over-reporting the favourable ones. In the case of employment status, individuals may be more inclined to over-report positive employment status and under-report unemployment duration or irregular

[11] In detail, Suvankulov (2010) uses German EOSP, Korea's KLIPS and US LFS and find a positive associate of 7% and 12.7% respectively more likely to being employed the next year. See for example Freeman, 2000; Bagues and Labini, 2007, Stevensen, 2008 and Suvankulov 2010.
[12] see Table 1.1 and 1.2 in Nakamura et al 2009

market behaviours. Third, there is no equivalence of data (same markets, with same time frequency) from the demand side to match and compare.

We follow the recent literature in the field, and utilise the power of computer programming in order to obtain an alternative dataset for understanding labour market conditions in Thailand. We turn to the data extracted from Thai's internet-based job portals and obtain two main types of datasets- the job ads and the resume posts. These datasets contain a number of desirable element. First, the datasets are instant reflection of the contemporary market at the time (as it is generated instantly by the users and can also be instantly extracted). Second, the data is obtained impersonally (not face-to-face) for self-report information in a real stake situation[13]. Thirdly, accessing these datasets is costless. And most importantly, the information from the demand side (vacancies posted by firms) and the supply side (resume posted by potential job seekers) are at the same time of access and broadly representing the same market. Therefore, this will allow us to take a close look at labour market tightness as well as other interactions within the labour markets which come from two sides of the same coin.

In this section, we will make use of two levels of data taken from internet-based job portals in Thailand. First, we will use to aggregate-level data for Thailand's biggest job portals to trace the time-varying trends of different characteristics of Thai labour markets. Second, we will turn to a micro, individual-level data to complement our analysis for Thai labour market at a point in time.

2.1 The aggregate data

The aggregate data we use is the publicly available data that the job portal website (Job Portal No. 1) has summarised on its pages. By locking on to the main pages, we can extract total number of job ads as well as of resumes presently on the site at the time of access. For the job ads, these numbers are also disaggregated by occupational categories (41), industry sectors (41) and locations (province-level). Similarly, numbers of resume are also grouped into occupations, fields of study (54), levels of qualification (3), age groups (5) and locations seeking work (province-level).

[13] Unlike survey questions, firms which post job ads on these job portals are committed financially to the cost charged by these job portals. For applicants, their probability of actual hiring depends strongly on the information they give to these websites. Therefore, they will be less likely to mis-report their personal information than in an equivalent survey question.

To obtain past information that no longer available on the present page, we use the service provided for free by an internet-archive web (http://archive.org/web). This internet-archive website allows us to "crawl" back to previous dates on a website of interest and collect any information that the internet-archive may have captured. For the Job Portal No. 1, we are able to crawl back and consistently collect relevant aggregate information from 2001 (the job portal website was firstly launched in 2000). Even if the dates for each year we crawled are not exactly matched, we are able to access and collect the data for at least once for each half of a calendar year. In sum, we are able to obtain the aggregate data on jobs and resume from the Job Portal No. 1 for 31 periods until August 2016.

2.2. The micro data

The micro-data we use here are the universe of unique job advertisements and of unique resumes of job searchers posted on a top online internet-based job portal (renamed as the Job Portal No. 2) during June 2016 to July 2016[14]. To collect the data, we made use of an automatic web crawling programme that searched separately for jobs and resumes posted on this website on a given period. As a result, we obtained all information of each job and each resume that are made publicly viewable. In details, there is a stark different between the nature of job advertising and resume posting. For firms, posting a job ads is costly. In general, a firm pay the website company for a bundle of service, including the duration of ads or the volume of ads and the total number of resumes the firm can view. (On average, the charge is of a minimum of one month of job-posting period.) In contrast, individual job seekers can put up their resume profile free of charge. Most job portals based in Thailand require the posting firm to also submit its official business registration in order to get verification. And all of the job portal website we have visited altogether bar job recruiting for direct sale, peer-to-peer marketing and insurance agents.

We collect the dataset at the individual job and individual resume level. Recall that our data is gathered from a public access data- therefore it does not contain any personal identification (especially the national identification number) or name of applicants. For the firm side, even though we do not have the firm's business registration number, we observe the name and address of these firms.

[14] When conducting a google search with the key word "looking for a job" (in Thai), this website is the top ranking site.

Note that in a given period our programme crawled the website, there exists jobs and resumes which had been posted before the dates of data access. For job ads, we also collect the date the job was last updated. Recall that a job post will normally remain in the database at least one month- even if the job may have been filled earlier. Therefore, our database of the job-post at the time of access will contain (a) jobs that still exist and (b) jobs that no longer exist but still within the period the firm has paid for. In total, we have 41,312 unique jobs ads during June-July 2016.

Since posting for a resume is free, our task to distinguish the active job-seekers from inactive ones is more challenging. To deal with this, we make use of the date at which a particular resume is last updated. In additionally, we acknowledge that there is a collection of resumes which are created but were not set as publicly searchable. Our database does not contain this type of resume. In total, we gather, within our chosen time period, over 80,000 resume for June-July 2016.

Further investigation (by reading contents on web forum) indicate that our chosen job portal websites represent a non-farming, formal, middle-to-low level sector of the labour market in the country. (Our further investigation, using web forum, indicates that there is a clear market segmentation among Thailand-based job portal websites- between the sites that match top-level jobs vs those that match middle-low level jobs.)

Occupation and industry categories are well indicated in both job ads and resume. The website has 73 tailored occupation categories as a drop-down option (with the addition of "others" and "intern/part-time" categories). Once matched with the International Standard of Occupation Code (2 digit-ISCO version 4, 2008), we aggregate these detailed occupations into 28 groups. As expected, the job ads do not contain any vacancies for agricultural-related (coded 9 of ISCO) and very few elementary service occupations (coded 8 of ISCO).

From the job ads, useful variables include number of vacancies per post (average 3 vacancies each), salary offered (47% of data), minimum work experience requirement (56%), education qualification (90%) as well as total number of views a job ads received between the date posted and the date we accessed the data.

Note that by law, firms based in Thailand are eligible to specify (a) gender, (b) age and/or (c) marital status of candidates they look for. (In fact, firms can also ask for military training background, fertility status or even a photograph of job applicants.). Our dataset also contains some of these insightful characteristic of Thai labour markets. Of all job ads, 75%

have target age range, 37% have gender-targeted (of which 14.7% target female applicants and 18% target males).

2.3. Data representativeness

As pointed out in Khun and Shen (2013), one can think of many reasons to suspect that our job portal sample is unrepresentative of the nation-wide actual vacancies and pool of job seekers. Despite the growth of the Internet and growing trend of digital-based economy, not all vacancies are captured in the internet-based job portals. We acknowledge that our database may not capture jobs which are (a) internally promoted, (b) relatively short vacancy spell, (c) upper-level and (d) of agricultural sector. In addition, even if we observe some jobs of the public sector, our sample under-represents vacancies from this sector[15].

Given that the pool of Internet users are younger, more educated and of course, more technologically apt than the population, our resume sample reflects the same story (Kuhn and Shen, 2013; Kurekova, 2014; Mang, 2012)[16]. To understand the degree of representativeness of our internet-based dataset (both the aggregate version and the micro-data), we map a number of characteristics from our sample with two nationally representative surveys conducted by the National Statistical Office.

To check for the representativeness of the job ads in our sample, we make use of the job vacancies reported in the 2013 Thailand Labour Demand of Establishment Survey (the LD thereafter). Note however that the LD only surveyed selected industries, which may not be the same as those recruiting via online tools. For the resume side, we cross-check our job portal sample with the Household Survey on Information and Communication Technology (ICT) of 2015 (ICT thereafter).

From Table 2.4, the ICT indicates that different age groups do not share the same online behaviours. Among those who use internet, the majority of individuals who use internet to help with job seeking activities concentrate among aged 20-29 (counted as 42% of those who use online tools for job-search). In the ICT, together, there are just over 1 million users who ever used internet for job searches. Our total count of resumes (total of 89,000 resumes updated within 2016) equals to 8.4% overall.

[15] In Kuhn and Shen (2013), they find that the pool of job ads from their Chinese job portal over-represented entre-level jobs, jobs in expanding, jobs with longer vacancy duration and of industries with high turn-over.

[16] As discussed earlier, Thailand's Internet penetration rate is 56 percent. Note also that the pool of resume in our sample will consists of three main types of individuals: those out of job, on-the-job searchers and recent graduates.

From Table 2.5-2.7, we find that, as mentioned in Khun and Shen (2013), our job ads over-represent higher skill candidates than the LD summary. Among different occupations, the comparison across different datasets indicates that our job portal database (micro-data and aggregate data) nearly mimics the exact number of total labour demand revealed in the LD survey for the Executives and the Professional occupations. However, in total, the sample of job ads from the internet falls significantly short of the total number of vacancies calculated in the LD. This is because the internet job portal extremely under-represent the demand for labour in low skilled occupations (namely agriculture, plant operators and services). We find this to be aligned with what suggest in the literature and that it is not a Thailand specific case.

III. Understanding two sides of the same coin with data from internet-based job portals

3.1 Tracking Thailand's labour market tightness from aggregate-level job portal database

Here, we start by presenting the aggregate labour tightness (defined as the ratio of vacancies to resumes) at the 1-digit occupation level. (Figure 3.1) Using the data of the Job Portal No.1 from the first half of 2012 to 2016 (August 2016), we notice that the ranking of labour tightness remains somewhat persistent across the period- with the service occupation category has the highest v/u and the clerk category places at the bottom. The trend is declining, due mainly to the fact that the vacancies are on the rise with a much smaller rate that the rise of job seekers.

At the regional level, the ratio of v/u from the lowest is North, North-east and South. And this is consistent across seven years of data. For these regions, the size of vacancies had remained very low, in comparison to the number of job seekers applying with the region. While the (reverse) labour tightness looked relative high for East between the years 2011-2012, its level dropped down thereafter. Largely, this decline stems from a slower rate of vacancy expansion than the size of job seekers in the East region. Figures 3.2A-3.2C plot regional levels of total job ads, total resumes and the measure of labour tightness over time.

More interestingly, when we examine the spatial distribution of labour tightness (Figure 3.3A), we notice that there might be a crude correlation of higher vacancies for provinces with better road transport access (seen in Figure 3.3B). To formally investigate this observation, we run a cross-sectional OLS, relating the aggregate labour tightness in 2016 at province level with a set of economic indicators, locational density of factories and infrastructure. The first specification takes the standardised labour market tightness as the outcome of interest ($Tightness_p$). The second specification assigns the percentage change of market tightness between as the outcome of interest ($\Delta Tightness_{p,\Delta t}$). We also use these specification to explore if the implementation of a uniform national statutory minimum wage of 300 Baht per day in January 2013 may pose any influence on the level and the change of labour market tightness locally (Lathapipat and Poggi 2016 and Lathapipat 2016)[17]. To do this, the variable $\Delta minwage_{p,2013}$ equals to percentage change of minimum wage pre-2013 and post-2013, specifically for each province, p.

$$Tightness_p = \boldsymbol{X}_p\boldsymbol{\alpha} + \Delta minwage_{p,2013} + \varepsilon_p \qquad \textbf{(Equation 1)}$$

$$\Delta Tightness_{p,\Delta t} = \boldsymbol{X}_p\boldsymbol{\alpha} + \Delta minwage_{p,2013} + \varepsilon_p \qquad \textbf{(Equation 2)}$$

Table 3.1 shows summary statistics of labour market tightness (job posts and resumes), economic measures (e.g. gross provincial product, Night Lights index, population size), transport infrastructures (road, rail) and change of minimum wage. Because the level of minimum wage in some provinces had already been adjusted upwards prior to the full implementation in January 2013, in our analysis, we try both $\Delta minwage_{2011,2013}$ as well as $\Delta minwage_{2012,2013}$. Column I shows statistics for provinces with labour market tightness in 2016 at least at median level and Column II is for the provinces of tightness level below the median.

Table 3.2 reports results from our regression analysis. Column I indicates that a percentage point change in minimum wage is related to a reduction of 0.3 standard deviation (sd.) in labour market tightness (measured in 2016). The significant relationship is, however, not detected when we regress the vacancy and the resume separately. The findings do not indicate much spatial influence of transports infrastructure on the level of v/u. Nevertheless,

[17] In their papers, Lathapipat and Poggi (2016) find that aggregate employment had been stable with some minor downward adjustments six quarters after the policy introduction for all types of workers, with no contraction in weekly working hours.

an existence of rail tracks in a province is associated to a decrease of 0.228 sd. of vacancy posting (Column III and IV). Column VII and VIII summarise the results from Equation 2. Again, we find that a percentage point rise of within-province minimum wage is related to a decline of the percentage change of tightness by 0.33 sd and 0.46 sd. (when measured against $\Delta minwage_{2011,2013}$ and $\Delta minwage_{2012,2013}$, respectively.

3.2. Learning about Thailand's labour market mismatch from micro-data job portal database

Now, we turn to our micro-data of a popular job portal. As explained in the previous section, we were able to access individual-level information from the site (Job Portal No.2). Without uncover any personal private identification, we extracted all job ads and all resumes (set for public accessibility) of the list of variables (shown in Table 2.2.A and 2.2.B).

Using information at this level, we begin by investigating if the high level of labour market tightness in some occupations (found in the last section) might be due to mismatch of skill requirements of jobs and potential applicants. Using this dataset, we can able to sub-category the job ads further-by education, gender, or desired years of work experience. Figure 3.4 shows the geographical distribution of skills. Crudely, we define applicants who obtain their qualification from technical vocation training, technically-focus universities (Rajamangala) and those with technical fields as those with hard degree. For those with commercial vocations or of other non-technical fields as with soft degree. Figure 3.4 compares the demand for applicants with bachelor or above from the job ads and the supply of job seekers with soft-degree. The comparison suggests that for most parts of the country, skills are located where they are demanded. However, when we compare Figure 3.4A and 3.4B to Figure 3.5C below, we notice that the provinces with high demand for college education are of those provinces with high market tightness (v/u).

In general, labour market friction is also believed to have cause by geographical mismatch of jobs available and applicants who are willing to relocate to a particular site. Note before that job ads in our database are heavily based in Bangkok (74% of total vacancies) whilst 61% of applicants are seeking jobs in Bangkok. Thus, we begin this examination by mapping the geographical distribution of jobs and separately of applicants across Thailand (based on their stated locational preference in the resume). Figure 3.5 shows that potential workforce do locate where most jobs are, with the exception of some lower North-Eastern

provinces. Nevertheless, the number of job seekers far outweighs total vacancies among popular locations. Therefore, when we plot the market tightness from our micro-data (Figure 3.4C), we observe that v/u is relatively low in these provinces. However, notice that along the eastern seaboard, our measure of market tightness looks relatively large.

Given that three quarter of total jobs are based in Bangkok, we look further if the geographical distribution of jobs and applicants, at the district level, display the same story within Bangkok. While the pool of Bangkok job seekers spread somewhat evenly across Bangkok 50 districts (with the exception of Ladprao, Bang Krapi, Bang-Na), job ads distribute much less uniformly. Bang Rak and Wattana have the highest (8% share) of total Bangkok job vacancies. Figure 3.6A-C displays the geographical spread of vacancies, applicants and market tightness around Bangkok. We can observe that within Bangkok, unfilled jobs locate either in the very central of the capital or around the outer districts. In contrast, preferred locations of work, indicated by applicants, concentrate more in the northern parts of the capital.

We list out two potential factors, which may contribute to the geographical mismatch of this selected Thai labour market. First, the presence of geographical immobility (that is job seeker's unwillingness to relocate) may account for unfilled vacancies in certain provinces. Second, within a province, a gap between the minimum accepted salary (the so-called reservation wage) and the maximum pays (the reservation productivity of workers) can potentially reduce the matching of applicants into jobs.

To support the first explanation, we take the location of the institution where the latest qualification was obtained as our proxy for the location of most current residence. We then map this location of residence to the location of work. Figure 3.7 presents the share of individual applicants' resident locations within each location choice of work. Whilst Bangkok can attract nearly 50% of applicants from outside its region, most provinces in the North and the Northeast have failed to reach 10%. This also means that the majority of applicants in the North and the Northeast are drawn from those who had studied in the regions.

To see if other observable individual characteristics may influence a choice of location for work, we run a regression analysis using our resume database. Let $Samesite_i$ be an indicator with value of 1 if the individual chooses her work location the same as her residence, $\boldsymbol{C}_i$ be a vector of individual characteristics observed in the dataset. Our mobility

regression is as below. We run a province-of-work fixed-effect model to account for unobservable variables at this level. The standard errors are also clustered at the region level.

$$Samesite_i = \boldsymbol{C}_i\boldsymbol{\gamma} + \mu_i \qquad \textbf{(Equation 3)}$$

The regression results are in Table 3.3. It shows that being a female job seeker is associated to 0.2% more likely to stay in the same area as the last education. The higher the years of work experience, the more likely a job seeker will stay put. Young job seekers are more geographical mobile. On average, individuals with higher grade point average are more likely to move away from where they previously studied.

Another source of labour mismatch can be due to the gap between the reservation wage and the expected labour productivity. Under a matching model, a worker would always reject the job offer if the offered wage is not at least equal to her reservation wage. Likewise, upon the initial match with a worker, the match will break if the worker is not as productive as the offered wage by the firm (Mortensen and Pissarides, 1994). Therefore, to investigate this potential mechanism, we look at salary differences within a well-defined job (matched job-applicant with the same level of qualification, experience and occupation demanded and supplied). Table 3.4 displays the average salary for each of these jobs.

We also run a simple Mincerian-style equation, separately for the demand side (job ads database) and the supply side (resume database). We do this to check if the applicants and firms share similarities in their wage-setting behaviour. In other words, we want to see if each additional skill or characteristic share the comparable market value for each side of the market.

Let W_j (W_i) be the log of level monthly-rate salary offered (requested) by a job ad j (an applicant i), $\boldsymbol{Z}_j$ ($\boldsymbol{Z}_i$) be a vector of individual job (applicant) characteristics observed in the dataset. Control variables include occupation categories and locations. The standard errors are also clustered at the region level. We derive Equation 4 for the job ads and the wage offer. Equation 5 represents the reservation wage equation, for the applicant side.

$$W_j = \boldsymbol{Z}_j\boldsymbol{\pi_j} + \varphi_j \qquad \textbf{(Equation 4)}$$

$$W_i = \boldsymbol{Z}_i \boldsymbol{\pi_i} + \rho_i \qquad \textbf{(Equation 5)}$$

We run a simple OLS Mincerian-style wage equation, each for the demand and the supply side of this particular labour market platform. By using log-form of monthly salary, all coefficients can now be interpreted as percentage points. Table 3.5.A displays the results with log of wage offer (at job ad level) as the dependent variable. Table 3.5.B presents the results with log of wage request (at individual resume level) as the dependent variable. In both equations, we controls for the same set of 1-digit occupation categories and province-of-job fixed effect. In both regressions, the standard errors are robust and clustered at the regional level.

From the wage offer side, it reveals that (under the full specification- Model III) job ads which target female only is related to just over 10 percent reduction in the wage offer. This penalty is not found for male-only job ads. Jobs seeking young workers look to offer smaller wage (18% negative for jobs targeting adolescents and 8% for jobs for aged 26-35). Work experience is valued by the demand side. One additional year of work experience is related to 10% increase in the wage offer. Jobs that do not specify a level of education qualification pays 6% more than jobs seeking bachelor graduate (bachelor degree is our omitted education category here). While bachelor degree or highest tops the salary level, jobs looking for the minimum education of middle school qualification (at present, the school-leaving minimum qualification) pays the least. Graduating with high school qualification is valued less than graduating with a high-level vocational school qualification in the job ads database.

Looking now at the reservation wage side, strikingly female applicants tend to ask for nearly 10% less wage than the males. Young job seekers (under 25 years) tend to set higher reservation wage than slightly older cohorts. However, the most senior age group is found to set the highest level of reservation wage in this database. Consistent with employers, job seekers do relate their work experience to higher requested level of pays. Without controlling for past salary (specification I), higher years of reported work experience is related to higher asking pay. However, once past salary is added, the magnitude of years of experience reduces around 50 percent. But overall, we observe a non-linear increase in the rise of experience and the rise of asking pays. In comparing to applicants seeking full-time jobs, those who look for

freelance do not reduce their asking wage. In contrast, those looking for internship or part-time job ask for roughly 25% less pay.

Interestingly, applicants value their school performance (proxy by Grade Point Average) as well as their self-report skills (English competency and other languages). With applicants of general bachelor degree as the education baseline, those with vocational qualifications (fewer years of schooling) request less pays. Notice that those of commercial VQ demand almost 3 times less pays than those of technical VQ. Among bachelor graduates, applicants with a Rajabhat degree is correlated with 7% lower wage reservation than the general degree. However, those with a Rajamangala on average set 6% higher wage.

IV: Is mismatch as a result of gender-targeting job opening?

Our argument is that because of many jobs tend to limit at the outlook, the gender of workers (be that discrimination or statistic learning), jobs are not filled and some jobs are over-subscribed. This, of course, causes disadvantages to the firms: unfilled vacancies are a productivity loss. And filling job with under-qualified but right gender also leads to productivity loss. For workers, it is a gain for those who not subjected to discrimination but loss to those who are. Furthermore, this affects the occupation paths and planning of future labour forces. Thus, national output is a risk.

From the dataset, we investigate on-the-surface, if this feature of gender-mismatch may take place in this sub-labour market. We begin by investigating the nature of gender-targeting job ads. By analysing a Chinese internet-based job portal, Kuhn and Shen (2013) find a *negative* correlation between the probability to gender-target their ads and the level of skills required in a job (measured by qualification level, work experience or occupational ranking). This suggests that the firm's strategy to gender-target goes hand in hand with its aim to skill-target.

From Table 4.1, it is plain to see that more than 30 percent of our sample of job ads (in the micro-data) is gender-target[18]. To investigate further, we group these job ads into different occupational categories, educational requirement, minimum work experience and

[18] Lawler and Bae (1998) looked at a sample of 902 job ads by MNCs in Thailand from the Bangkok Post (English language newspaper) and checked for gender-targeting behaviour of big corporations. They found that 25.2% were male-only, 13.9% were female only and 60.3% were gender-non-specific. Among only Thai firms, 32.5% were male-only, 13.1 were female only.

age-group of ideal candidates. Figure 3.8 plots the share of job ads by its gender target and 1-digit occupation code. By ranking from most highly skilled occupation (executives) to least skilled occupation (services), we observe that the share of gender-neutral job ads gets smaller as the skill requirement declines. We check with the 2013 Labour Demand Survey to see if the corresponding stylise is found. Figure 3.9 reflects a consistent story between our two samples.

More strikingly, as education requirement gets higher (ranking from minimum requirement of lower secondary school to master degree), we notice a rise of gender-neutral job ads (Figure 3.10). Again, we find the similar but weaker correlation when we plot the share of gender-target job ads with work experience.

However, when we examine the gender-skill targeting relationship using age requirement as a proxy (as age is generally translated to experience), we find the contrary (Figure 3.11). Job ads that aim at young candidates (aged 15-19) are less gender-specific that job ads for those of 25 years and above. Instead, among those gender-target job ads that aim for older candidates, a higher proportion of jobs target male candidates more heavily (from 23 percent for aged 20-24 to 27 percent for aged 30 or above). These leave much room for interpretation. Among the pool of younger candidates, both males and female are equally less likely to have a family. As the pool gets older, females become less ideal workers to a firm. Hence, job openings which aim to recruit older workers are more likely to target and screen for males.

V. Summary and conclusions

In the era of the digital economy, technology has become a part of our everyday lives. The way we do our economic activities has changed from how we did them in the traditional economy. The labour market in the digital age is also different from the traditional labour market. Specifically, both employers and job applicants rely more on the internet when posting vacancies, applying for jobs, and searching for information relating to employment benefits. These internet activities create the digital footprints that can be used to analyse the labour market in the way that could not be done by utilising the traditional labour data.

The objective of this study is to use the internet-based data (e.g. Google Trends data, and the internet job board data) to analyse the labour market from a new perspective. Our study demonstrates that: (i) the data from the internet job boards allow us to understand the labour market from both the demand side (firms' vacancy postings) and the supply side (applicants' resume postings). In addition, we can also observe the distribution of jobs and workers across space and study the spatial concentration of jobs and workers of specific skills and occupations. (ii) This firm-applicant dataset collected from the internet job boards also allow us to study the disparity between the offered wage (by firms) and the asking wage (by applicants) across types of jobs, locations, and skills. (iii) And with careful analysis of the internet job board data, we can extract a proxy for labour market tightness at the cross section of job characteristics, across space, and over time.

The use of internet-based data to analyse the labour market may still be uncommon in Thailand. And the access to such data may still be limited. However, we believe that these data provide new meaningful insights for the Thai labour market. Therefore, it could be helpful for the policymakers to pay attention to these data along with the traditional labour data when making policy decisions.

References

Antenucci, D., Cafarella, M., Levenstein, M. C., Re, C., & Shapiro, M. D. (2014, March). Using Social Media to Measure Labor Market Flows. NBER Working Paper No. 20010.

Askitas, N., & Zimmermann, K. F. (2009, June). Google Econometrics and Unemployment Forecasting. IZA Discussion Paper Series No. 4201.

Autor, D. H. (2001). Wiring the labor market. *The Journal of Economic Perspectives*, *15*(1), 25-40.

Barnichon, R. (2010). Building a composite help-wanted index. Economics Letters, 109(3), 175-178.

Barnichon, R., & Figura, A. (2015). Labor market heterogeneity and the aggregate matching function. *American Economic Journal: Macroeconomics*, *7*(4), 222-249.

Choi, C., & Varian, H. (2009a, April). Predicting the Present with Google Trends. Google Inc.

Deming, D., & Kahn, L. (2015). Firm Heterogeneity in Skill Demands. NBER Chapters.

Diamond, P. A., & Şahin, A. (2015). Shifts in the Beveridge curve. *Research in Economics*, *69*(1), 18-25.

Edelman, B (2012) "Using internet data for economic research" J Economic Perspectives

Fabo and Tijdens (2014) "Using web data to measure the demand for skills" CELSI Discussion paper no. 21.

Hershbein, B. J., & Kahn, L. B. (2016). Do Recessions Accelerate Routine-Biased Technological Change? Evidence from Vacancy Postings.

Khun, P., & Shen, K. (2013). Gender Discrimination in Job Ads: Evidence from China. Quarterly Journal of Economics, 128(1), 287-336.

Krueger (2000) "The internet is lowering the cost of advertising and searching for jobs" New York Times, July 20th.

Kuhn, P., & Mansour, H. (2014). Is Internet job search still ineffective?. *The Economic Journal*, 124(581), 1213-1233.

Kuhn, P., & Skuterud, M. (2004). Internet job search and unemployment durations. *The American Economic Review*, *94*(1), 218-232.

Kurekova, L. (2014) “Using Internet Data to Analyse the Labour Market: A Methodological Enquiry” (IZA)

Lawler, J. J., & Bae, J. (1998). Overt employment discrimination by multinational firms: Cultural and economic influences in a developing country. *Industrial Relations: A Journal of Economy and Society*, *37*(2), 126-152.

Lenaerts, K., Beblavy, M., & Fabo, B. (2016). Prospects for Utilisation of Non-Vacancy Internet Data in Labour Market Analysis - An Overview. IZA Journal of Labor Economics, 5(1).

Lathapipat, D. (2016): “Assessing the effects of the minimum wage on wage inequality: An extension of the Unconditional Quantile Regression to a Provincial Panel.” PIER Discussion Paper

Lathapipat, D., & Poggi, C. (2016). From Many to One: Minimum Wage Effects in Thailand (No. 41.). PIER Discussion Paper

Marinescu, I. (2016, July). The General Equilibrium Impacts of Unemployment Insurance: Evidence from a Large Online Job Board. NBER Working Paper No. 22447.

Mortensen, D. T., & Pissarides, C. A. (1994). Job creation and job destruction in the theory of unemployment. *The review of economic studies*, *61*(3), 397-415.

Nakamura et al (2010) “Jobs Online”

Sahin, Aysegul, Joseph Song, Giorgio Topa, and Giovanni L. Violante. "Measuring mismatch in the US labor market." Manuscript, revised March (2011).

Shimer, R. (2005). The cyclical behavior of equilibrium unemployment and vacancies. *The American Economic Review*, *95*(1), 25-49.

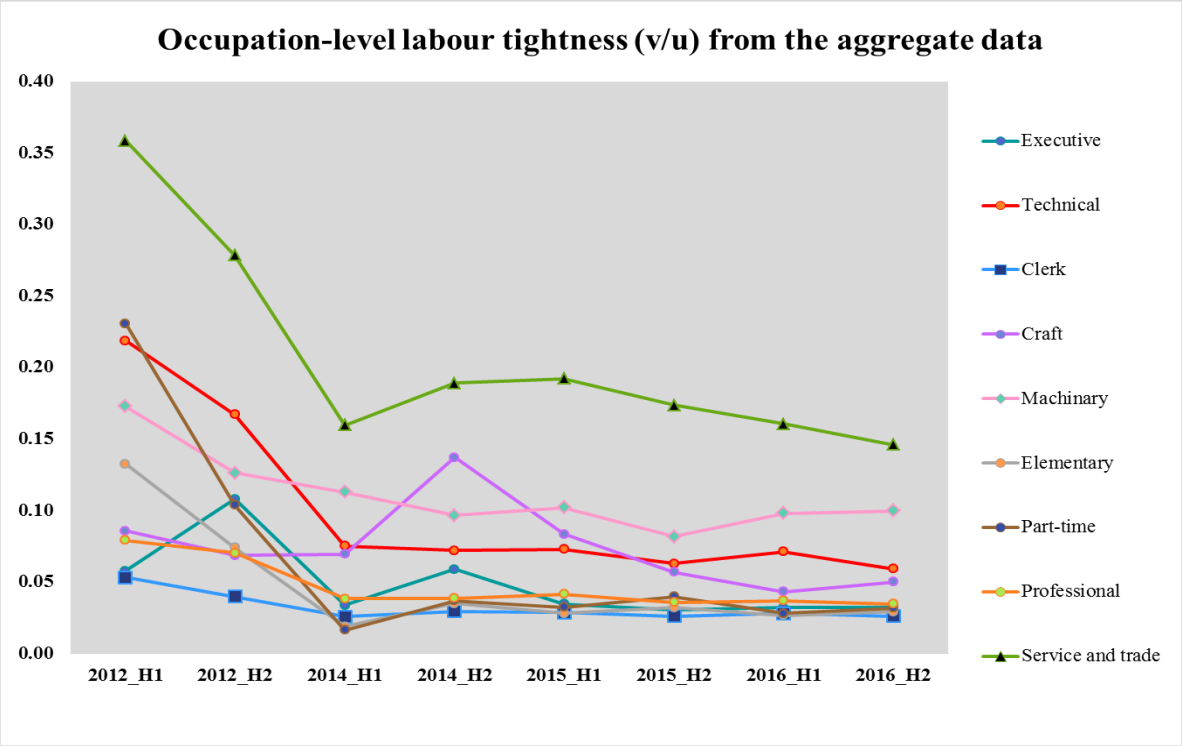
Stevenson, B. (2009). The Internet and Job Search. In *Studies of Labor Market Intermediation* (pp. 67-86). University of Chicago Press.

http://burning-glass.com/

https://www.conference-board.org/data/helpwantedonline.cfm

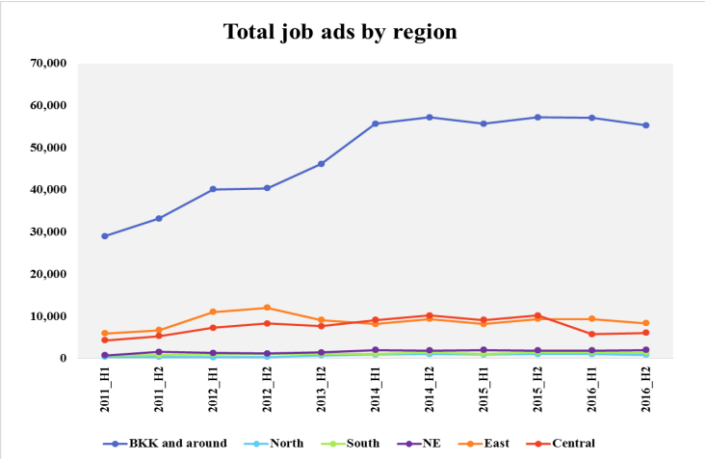
Figures

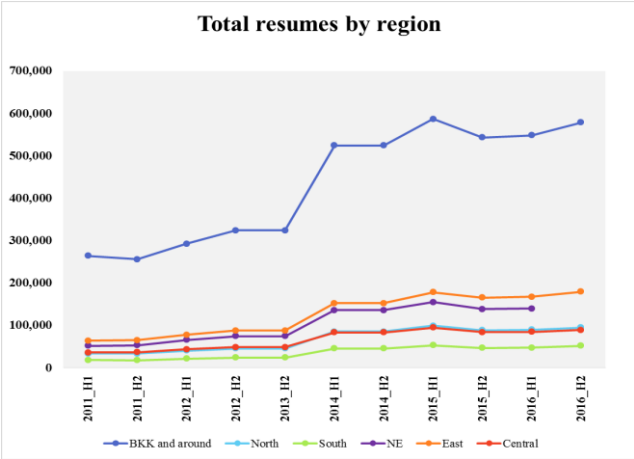
Figure 3.1: Trends of labour tightness by occupation

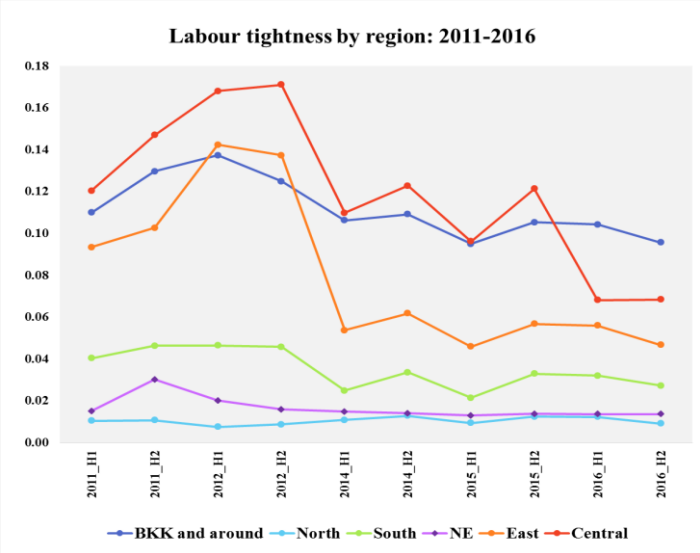


Source: Archived data of the Job Portal No.1

Figure 3.2A-3.2C: Total job ads, total resume and the measure of labour tightness over time by region

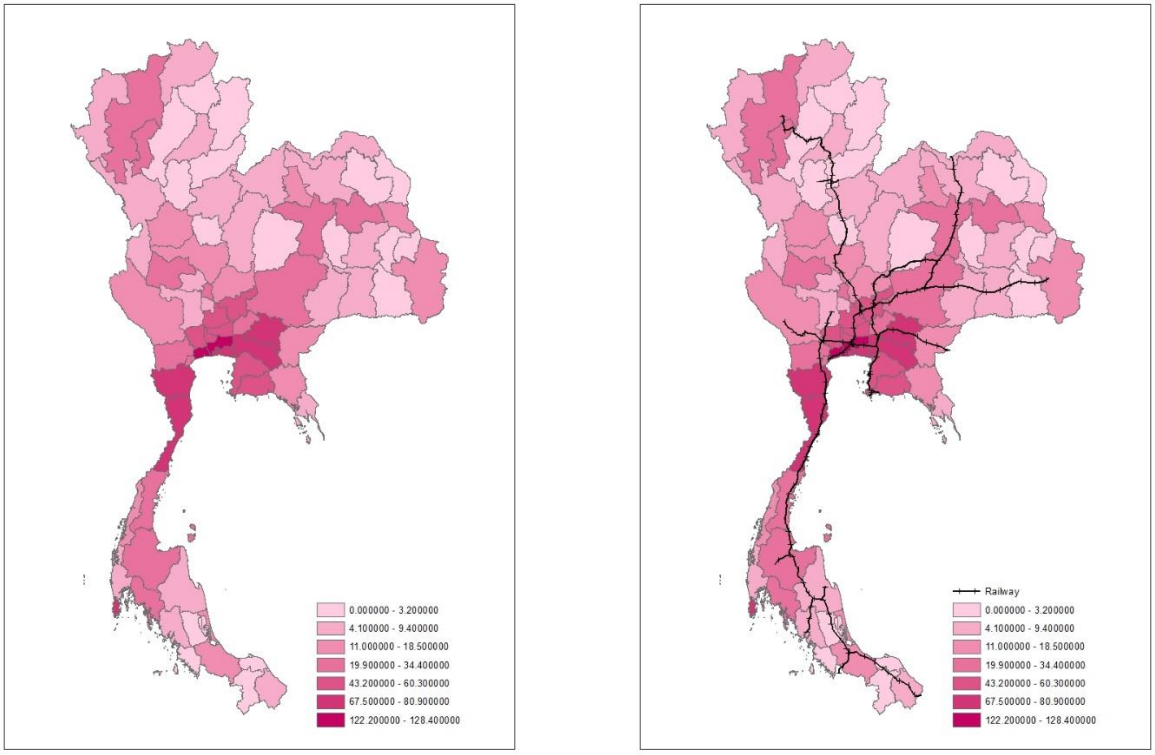






Source: Archived data of the Job Portal No.1

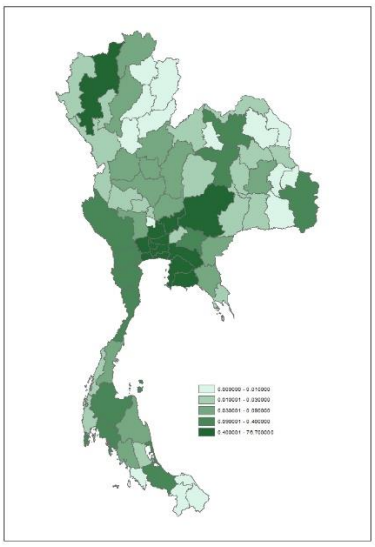
Figure 3.3A-B: Geographical distribution of labour market tightness with railways overlaid. (Source: aggregate data from Job Portal No.1, 2016)



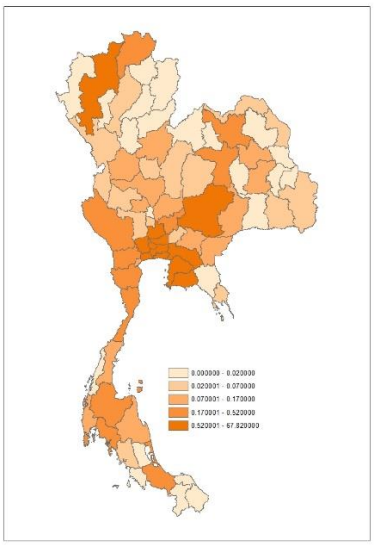
Sources: Archived data of the Job Portal No.1 (2016) and Ministry of Transports (2010)

Figure 3.4A-E: Distribution of demand for bachelor degree or above, compare to the supply of job seekers with the same qualification (as percentage of total jobs or resumes within each province). Source: The Job Portal No.2 (2016)

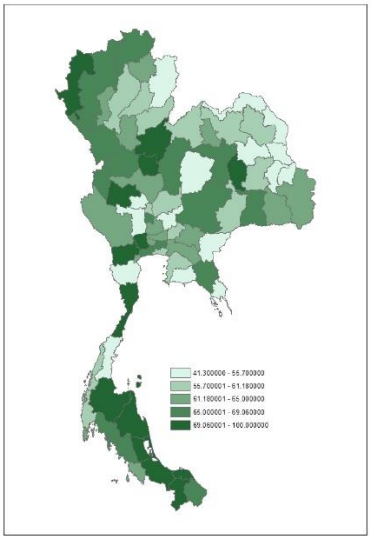
(A) Demand for bachelor degree



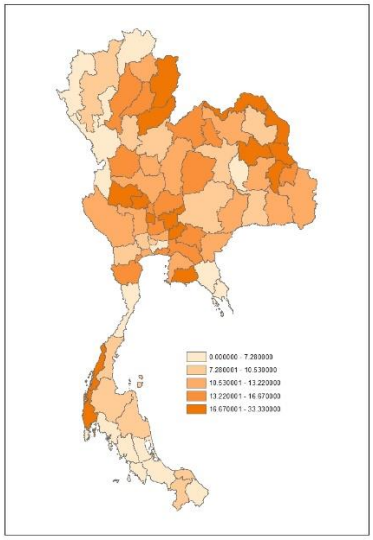
(B) Demand for high vocational degree



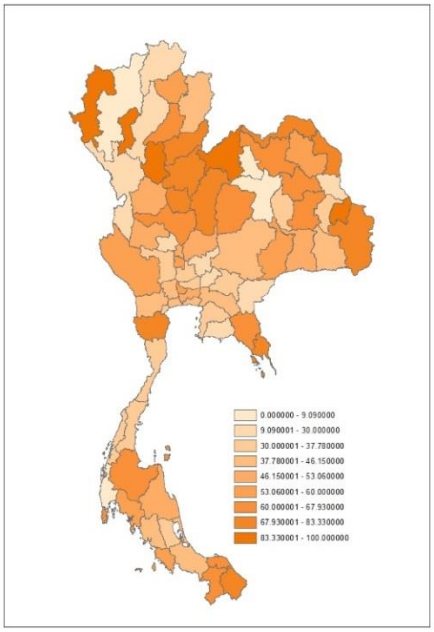
(C) Supply of bachelor degree



(D) Supply of high vocational degree



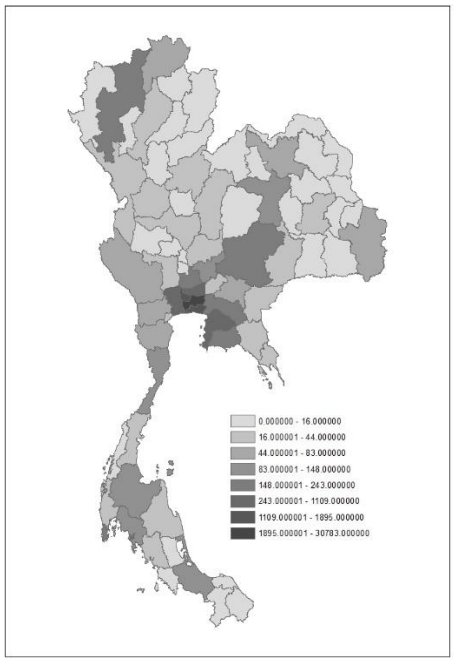
(E) Supply of soft-degree (% of total)



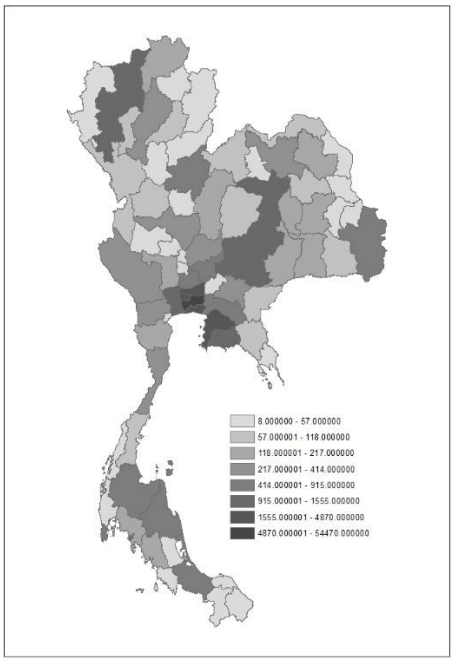
Source: The Job Portal No.2 (2016)

Figure 3.5A-C: Distribution of vacancies, resumes and market tightness around Thailand provinces. (Source: micro-data from The Job Portal No.2, 2016)

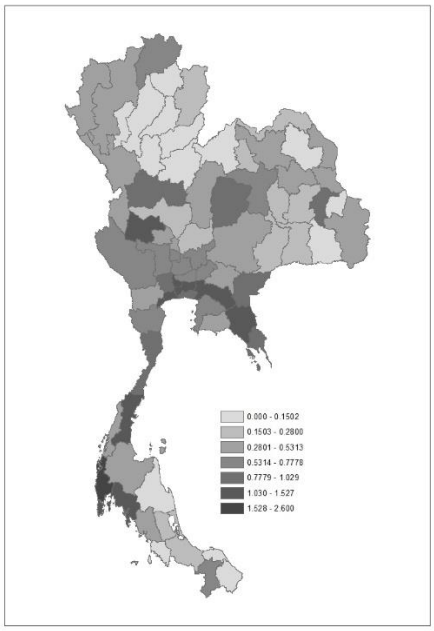
(A) Distribution of total vacancies



(B) Distribution of total resumes



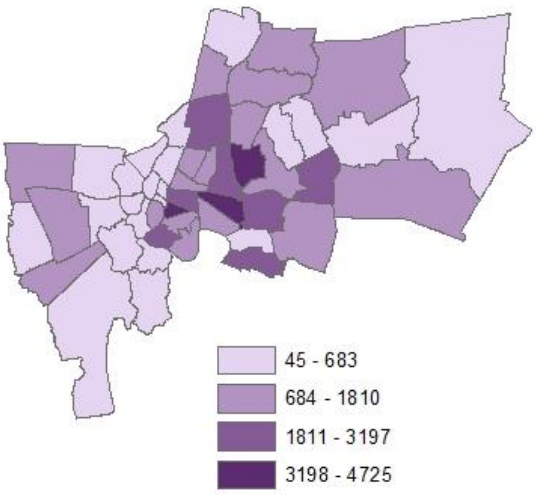
(C) Distribution of market tightness



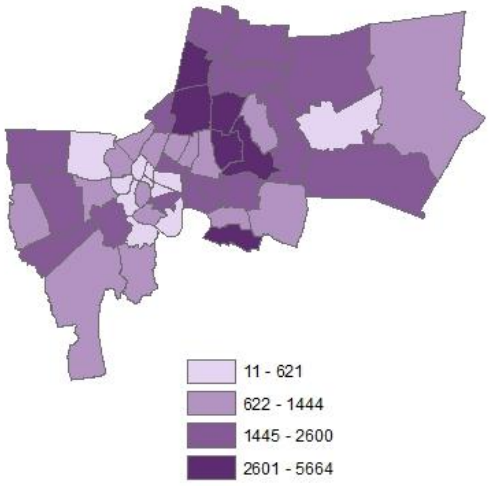
Source: The Job Portal No.2 (2016)

Figure 3.6A-C: Distribution of vacancies, resumes and market tightness (v/u) around Bangkok districts

(A) Share of total vacancies around Bangkok



(B) Share of total resumes around Bangkok



(C) Distribution of market tightness

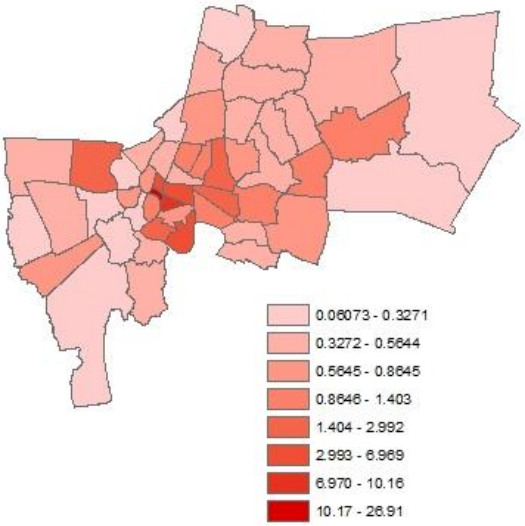
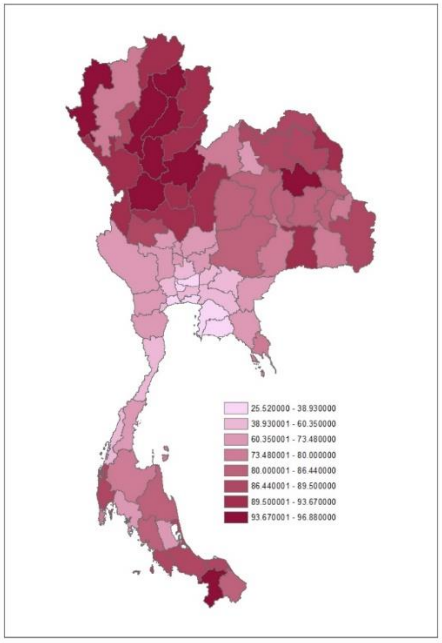
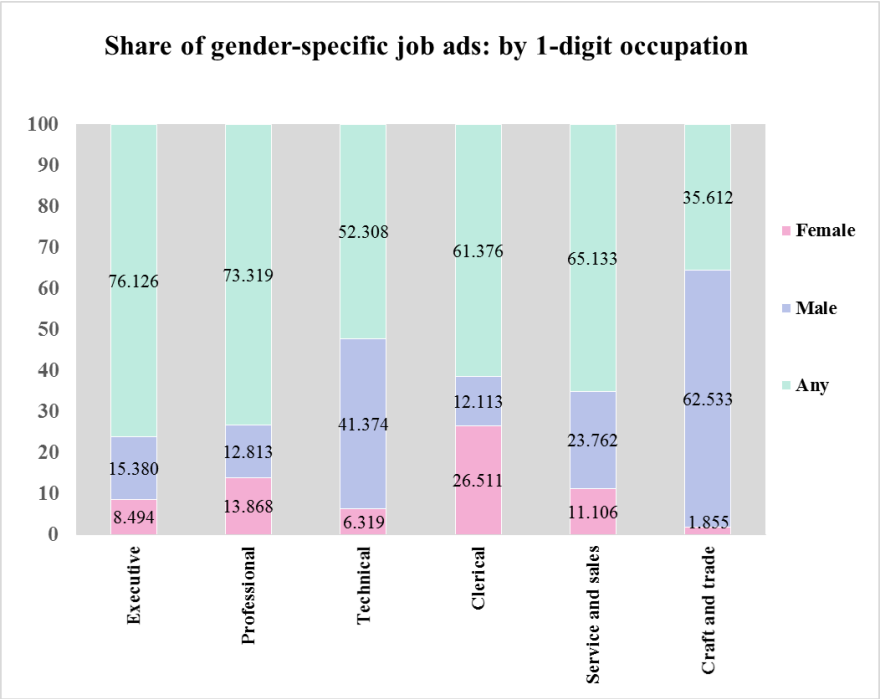


Figure 3.7: Share of applicants from the same region of education location in a chosen province of work



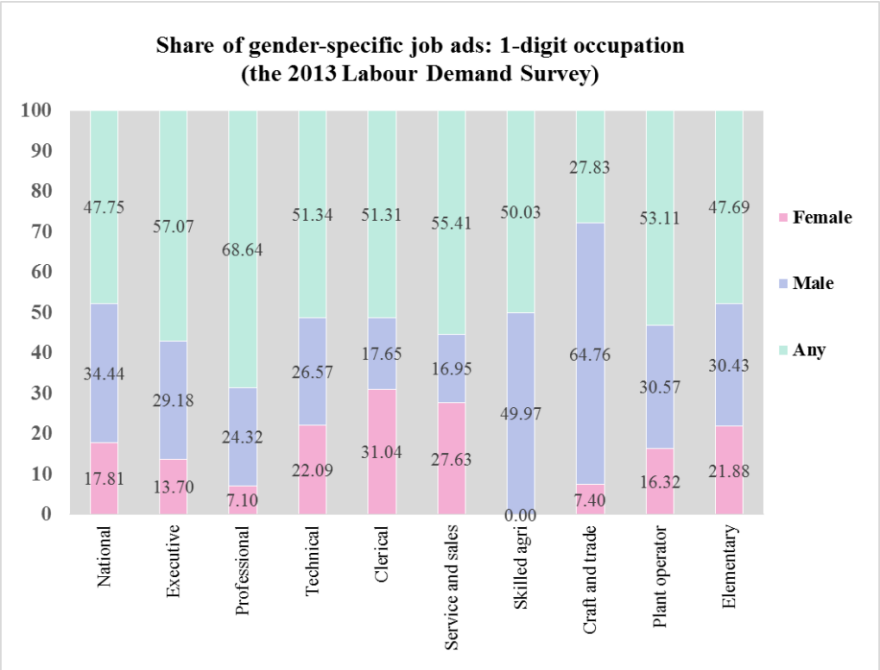
Source: the Job Portal No.2 (2016). The original region is defined as the region of the institution where an applicant obtained the latest qualification

Figure 3.8: The share of job ads by its gender target and 1-digit occupation code.



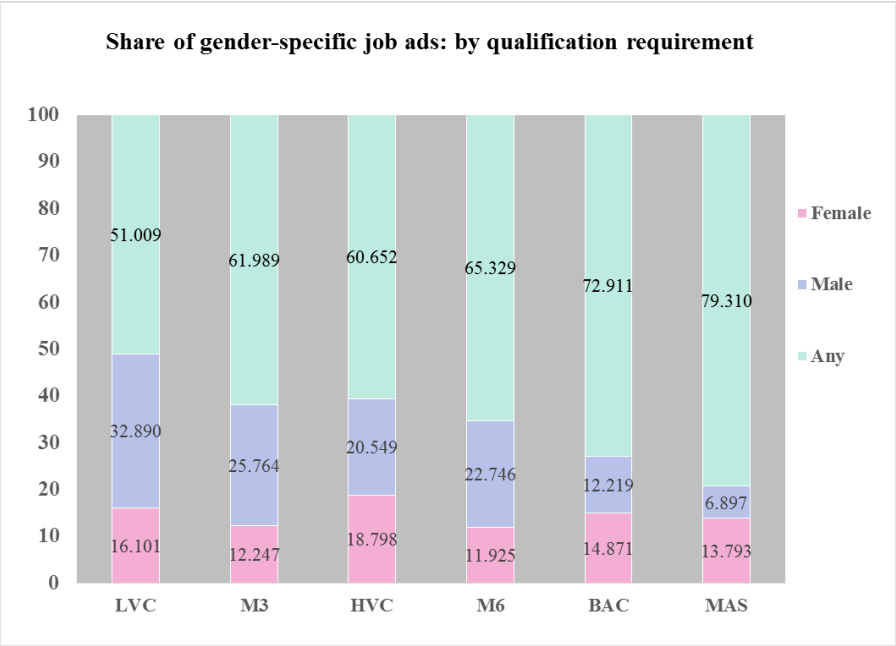
Source: The Job Portal No.2 (2016).

Figure 3.9: The share of job ads by its gender target and 1-digit occupation code from the LD survey



Source: The Labour Demand of Establishment Survey (2013).

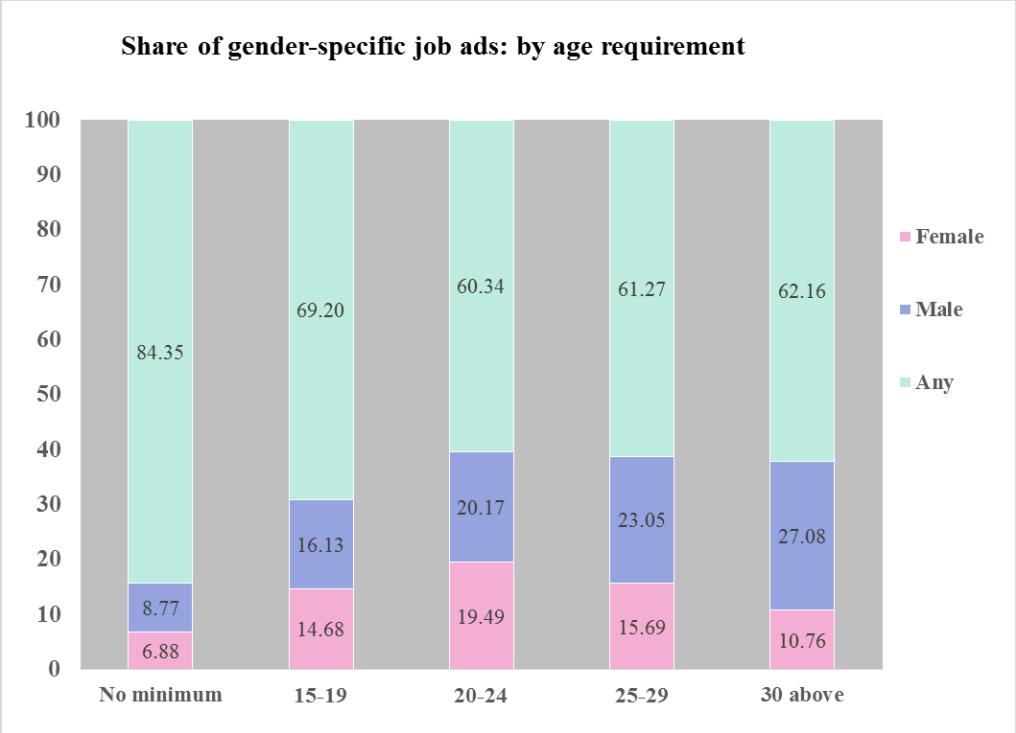
Figure 3.10A-B: The share of job ads by its gender target and skills requirements (qualification and work experience)



Source: The Job Portal No.2 (2016).

Source: The Job Portal No.2 (2016).

Figure 3.11: The share of job ads by its gender target and age requirements



Tables

Table 1.1: Share of job search methods from the ICT Survey 2015

	Total	Penetration rate of internet as job search platform
Looking for job or sending/submitting job application (aged 15 and older)	1,062,208	(%)
Total of employed (LFS 2016 Q1)	37,000,000	2.87
Total working age	55,000,000	1.91
Total unemployed (LFS 2016 Q1)	370,000	287.08
Total under-employed (LFS 2016 Q1)	237,500	447.25
Total under-employ & unemployed (LFS2016 Q1)	607,500	174.85

Sources: LFS 2016 Q1 and ICT Survey 2015

Table 2.1: Variables available from the Job Portal No.1 (aggregate data)

	Variables	Years	Num. categories
Job ads			
	Occupation	2001-2016	41
	Industry	2001-2016	41
	Region	2001-2016	6
	Province	2009-2016	77
Resumes			
	Occupation	2012-2016	41
	Field of study	2001-2016	44
	Qualification	2001-2016	3
	Age	2001-2016	5

Source: the Job Portal No.1

Table 2.2A: Variable descriptions and summary statistics of job ads from the Job Portal No.2 (micro-data)

Job ads (Total observations: 41,312)		
Variables	**Values**	**% non-missing**
Occupation	73	0.00
Region	6	0.01
Province	76	0.01
Salary offer (bahts)	Mean of 20,315	47.26
Work experience	0-5 years	56.73
Qualification	Lower vocation to PhD	90.61
Num.vacancies	Mean of 2.44	4.28

Source: the Job Portal No.2 (micro-data)

Table 2.2B: Variable descriptions and summary statistics of resume posts from the Job Portal No.2 (micro-data)

Resumes (Total observations: 89,313)		
Variables	**Values**	**% Non-missing**
Province (for jobs)	76	0.043
Gender	Male, Female	0.008
Age	Mean of 28.5	0.000
Reported work experience (Yrs)	Mean of 4.17	0.000
Chosen start date	4	0.303
Requested salary (Bahts)	Mean of 16,500	99.366
Occupation	68	0.645
Qualification	Middle School 3 to PhD	10.166
Work status	FT, PT, Freelance, Intern	0.000
Grade point average (out of 4)	Mean of 2.74	15.222
Last salary	Mean of 15,110	21.868
Province (of education)	76	58.539
Type of edu institution	10	12.189
Language ability	Score 1-3	18.791

Source: the Job Portal No.2 (micro-data)

Table 2.3: Share of job ads with and without targeting characteristics of job candidates

Variables	Num. categories	% Non-missing
Gender target	male, female, both	0.00
Age target	15 years minimum	74.72

Source: the Job Portal No.2 (micro-data)

Table 2.4: Comparing the characteristics of the Job Portal's resume data and the ICT survey

	15-19	20-24	25-29	30-34	35-39	40-49	50- 59	60 +	Total
(I) Individual used internet-base job search ever (ICT survey)	107,248	335,594	272,721	158,085	99,266	67,523	17,288	4,483	1,062,208
(II) Share of total users of internet-base job search by age (I/ Total)	10.10	31.59	25.67	14.88	9.35	6.36	1.63	0.42	100
(III) % of working age pop (ICT survey)	12.19	10.49	6.54	3.67	2.14	0.71	0.23	0.12	
(IV) Total Job portal resume database (in 2016)	1,899	25,479	28,436	19,266	9,419	4307	486	16	89,308
(V) Share of resume database by age (IV/Total)	2.126	28.529	31.840	21.573	10.547	4.823	0.544	0.018	100
(VI) % of resume to total ever users (IV / I)	1.771	7.592	10.427	12.187	9.489	6.379	2.811	0.357	8.408

Sources: The 2015 ICT survey (NSO), LFS 2016 Quarter 1 (NSO), job portal (accessed June/July 2016- all resumes with last update from Jan 2016-July 2016)

Table 2.5: Occupational distribution of the Job Portal's job ads data and the LD survey

ISCO-08 1 Digit	Occupations	Vacancy Survey	Job portal No.2 micro-data	Job portal No 1	
		2013	2016	2013	2016
1	Managers, senior officials and legislators	0.59	3.90	1.24	1.03
2	Professionals	7.68	59.20	38.05	39.36
3	Technicians and associate professionals	7.80	4.57	12.10	10.08
4	Clerical support workers	6.82	17.35	17.49	17.71
5	Service and sales workers	18.58	10.03	26.01	26.89
6	Skilled agricultural, forestry and fishery workers	0.26	0.00	0.00	0.00
7	Craft and related trades workers	23.53	4.73	0.14	0.13
8	Plant and machine operators, and assemblers	20.25	0.00	2.79	2.87
9	Elementary occupations	14.49	0.22	2.17	1.94

Sources: The 2013 Labour Demand Survey (NSO), micro-data Job Portal No.2, aggregated Job Portal No.1 archives

Table 2.6: Regional distribution of the Job Portal's job ads data and the LD survey

	LD Survey	Job portal No.2	Job portal No. 1	
	2013	2016	2013	2016
Bangkok	54.89	85.38	69.70	74.69
Northern	9.78	1.24	1.11	1.17
Southern	9.08	1.95	1.70	1.90
Northeastern	10.33	1.86	2.15	2.69
Central	15.91	9.57	25.34	19.54

Sources: The 2013 Labour Demand Survey (NSO), micro-data Job Portal No.2, aggregated Job Portal No.1 archives

Table 2.7: Educational distribution of the Job Portal's job ads data and the LD survey

	LD Survey	Job portal No.2
	2013	2016
Lower secondary or lower	32.83	16.64
Upper secondary, Vocational	18.84	26.80
Post-secondary, Higher vocational	8.58	13.63
Bachelor degree or higher	15.02	35.86
Not specify	24.73	7.08

Sources: The 2013 Labour Demand Survey (NSO), micro-data Job Portal No.2, aggregated Job Portal No.1 archives

Table 3.1: Summary statistics of province-level labour tightness, minimum wage and indicators

	Upper range v/u		Lower range v/u	
	Mean	**[SD]**	**Mean**	**[SD]**
	(I)		**(II)**	
v/u in 2016	**0.056**	**[0.029]**	**0.008**	**[0.006]**
% Minimum wage between 2011 and 2013	58.0	[14.7]	77.9	[6.57]
% Minimum wage between 2012 and 2013	13.4	[10.3]	27.5	[4.71]
Population Density	0.852	[1.54]	0.265	[0.35]
Share of inwards migrants	0.055	[0.101]	0.012	[0.018]
Average workers in factories	141,722	[156760]	15,186	[14802]
Incidence of rail tracks	0.826	[0.38]	0.519	[0.504]
Density of major roads (2010)	0.290	[0.33]	0.146	[0.25]
Density of minor road (2010)	1.887	[1.73]	1.345	[1.204]
GPP per cap (2014)	271,433	[198988]	89,833	[36935]
Debt-loan ratio of commercial banks	75.8	[25.83]	100.2	[30.44]
Share of big-size firms	2.6	[1.66]	0.9	[0.35]
Share of workers in big-size firms	48.6	[13.87]	27.2	[9.26]
Number of factories	3,147.00	[3937]	1,275	[1066]
Night Lights Index (2013)	24.8	[20.06]	5.7	[4.95]

Sources: Job Portal No.1Ministry of Labour, Bank of Thailand, National Statistical Office, Ministry of Interior, Ministry of Transports, NOAA (US).

Table 3.2: Relationship between labour market tightness and the 300-Baht minimum wage policy

	Tightness-2016		Vacancies-2016		Resume-2016		Change of tightness (2016,2011)	Change of tightness (2016,2012)
	(I)	(II)	(III)	(IV)	(V)	(VI)	(VII)	(VIII)
% Minimum wage between 2011 and 2013	-0.030***		0.002		0.004		-0.033*	
	[0.007]		[0.010]		[0.008]		[0.018]	
% Minimum wage between 2012 and 2013		-0.042***		0.001		0.006		-0.064**
		[0.010]		[0.014]		[0.012]		[0.026]
Incidence of rail tracks	-0.004	-0.002	-0.228*	-0.228*	-0.084	-0.085	0.177	-0.548*
	[0.158]	[0.157]	[0.096]	[0.096]	[0.065]	[0.066]	[0.286]	[0.284]
Density of major roads (2010)	-0.226	-0.215	0.4	0.403	0.363	0.364	-0.084	-0.055
	[0.161]	[0.159]	[0.262]	[0.260]	[0.252]	[0.251]	[0.539]	[0.534]
Density of minor road (2010)	-0.091	-0.094	0.080*	0.08	0.092*	0.092*	-0.065	0.021
	[0.064]	[0.064]	[0.041]	[0.042]	[0.044]	[0.044]	[0.113]	[0.112]

Notes: Robust standard errors in brackets. *** p<0.01, ** p<0.05, * p<0.1. All dependent variables are standardised. All equations includes additional controls of population density, share of inwards migration, number of workers in production factories, GPP per capita (2013), Night light index (2013), share of big enterprises, number of production factors and commercial banks' debt-loan ratio (2016). All regressions are clustered at region.

Table 3.3: Probability of staying at the same province

Dependent variables:	Same province of work and resident			Same region of work and residence		
Female	0.023***	0.022***	0.024***	0.022***	0.022***	0.024***
	[0.003]	[0.003]	[0.004]	[0.003]	[0.003]	[0.004]
Age 26-35	-0.044***	-0.047***	-0.052***	-0.053***	-0.052***	-0.055***
	[0.004]	[0.005]	[0.006]	[0.004]	[0.005]	[0.006]
Age 36-45	0.001	-0.024***	-0.030***	-0.025***	-0.040***	-0.044***
	[0.005]	[0.007]	[0.008]	[0.005]	[0.007]	[0.008]
Age 46-55	0.067***	0.018	0.015	0.035***	0.004	0.001
	[0.011]	[0.013]	[0.014]	[0.012]	[0.014]	[0.015]
Age 56 and above	0.095**	0.003	-0.014	0.048	-0.011	-0.022
	[0.043]	[0.047]	[0.050]	[0.045]	[0.048]	[0.051]
Currently working		0	0.001		0.004	0.004
		[0.004]	[0.004]		[0.004]	[0.005]
Experience 1-4 yrs		0.015***	0.015***		0.009*	0.009*
		[0.005]	[0.005]		[0.004]	[0.005]
Experience 5-10 yrs		0.022***	0.024***		0.008	0.007
		[0.005]	[0.006]		[0.005]	[0.006]
Experience 11-20 yrs		0.053***	0.052***		0.037***	0.034***
		[0.008]	[0.008]		[0.008]	[0.008]
Experience 21 yrs and more		0.069***	0.065***		0.038**	0.033*
		[0.018]	[0.019]		[0.018]	[0.019]
Want freelance job		0.072***	0.077***		0.057***	0.067***
		[0.020]	[0.021]		[0.020]	[0.021]
Want intern job		0.053	0.066*		0.060**	0.079**
		[0.034]	[0.037]		[0.030]	[0.034]
Want part-time job		0.052***	0.054***		0.041***	0.044***
		[0.012]	[0.013]		[0.011]	[0.013]
Currently studying		0.080***	0.130***		0.042	0.092**
		[0.031]	[0.042]		[0.030]	[0.042]
GPA		-0.015***	-0.014***		-0.014***	-0.013***
		[0.004]	[0.004]		[0.004]	[0.004]
Technical vocational		-0.006	-0.006		-0.013	-0.011
		[0.010]	[0.011]		[0.010]	[0.011]
Commercial vocational		0.137***	0.140***		0.102***	0.109***
		[0.009]	[0.010]		[0.009]	[0.010]
Rajamakala bachelor		0.022***	0.025***		0.016***	0.018***
		[0.004]	[0.005]		[0.005]	[0.005]
Rajabhat bachelor		0.020***	0.023***		0.035***	0.039***
		[0.004]	[0.005]		[0.004]	[0.005]
General edu school		0.114***	0.112***		0.093***	0.092***
		[0.012]	[0.013]		[0.012]	[0.013]
Adult learning		0.179***	0.183***		0.166***	0.173***
		[0.020]	[0.023]		[0.018]	[0.022]
Skill level in English			0.006*			0.001
			[0.003]			[0.003]
Observations	94,949	89,895	77,475	90,553	85,415	72,697
Adjusted R-squared	0.072	0.086	0.087	0.124	0.135	0.123

Notes: ***1%, ** 5%, * 10% clustered standard errors. Other control variables are education qualification and province of work fixed effect. The data is taken from micro-data resume posting, accessed in July 2016. Omitted categories are those aged under 15, bachelor degree and no work experience.

Table 3.4 Average salary by occupation from the job-seeker's reservation level and the firm's offered level

Occupation	Reservation wage		Wage offer	
	Mean	(SD)	Mean	(SD)
Executive	23502.13	(18260.61)	33799.93	(19084.39)
Professional	21240.78	(11208.51)	22487.45	(12935.98)
Technical	19075.07	(8160.46)	17864.9	(8731.87)
Clerical	16321.66	(7038.95)	16985.58	(8763.45)
Service and sales	16503.21	(8368.79)	13942.17	(5244.74)
Craft and trade	18000.42	(9476.92)	16038.65	(8272.37)
Elementary	15568.18	(4048.48)	17681.82	(12693.08)
Average	19525.79	(10877.72)	20315.56	(12332.64)

Source: micro-data Job Portal No.2

Table 3.5.A: Regression for wage offer by firms

Dependent variable: Log of wage offer		
Female only job	-0.046***	-0.047***
	[0.007]	[0.006]
Male only	-0.092***	-0.002
	[0.008]	[0.006]
Max age 26-30	0.082***	0.078***
	[0.017]	[0.012]
Max age over 31	0.152***	0.163***
	[0.017]	[0.013]
With max age target	-0.110***	-0.038***
	[0.007]	[0.005]
With experience target		-0.162***
		[0.007]
Years of experience		0.150***
		[0.004]
Num. vacancies		-0.007***
		[0.001]
Num. views		-0.000**
		[0.000]
With edu target		-0.032**
		[0.012]
LVC		-0.363***
		[0.008]
M3		-0.471***
		[0.008]
HVC		-0.268***
		[0.007]
M6		-0.447***
		[0.008]
MAS		0.212***
		[0.055]
Manager		0.225***
		[0.017]
Technical		-0.083***
		[0.011]
Clerical Support		-0.133***
		[0.006]
Services and Sales		-0.139***
		[0.007]
Craft and Related Trades		-0.094***
		[0.011]
Elementary		-0.156***
		[0.053]
Others		-0.220***
		[0.014]
Observations	21,311	19,689
Adjusted R-squared	0.055	0.507

Notes: ***1%, ** 5%, * 10% clustered standard errors. Other control variables are occupations and province of work fixed effect. The data is taken from micro-data resume posting, accessed in July 2016. Omitted categories are jobs omitted categories are gender-nonspecific jobs, professional occupation, bachelor degree and jobs require applicants' age under 25 yrs. All coefficients are of percentage point unit.

Table 3.5.B: Regression for wage reservations stated by job seekers

Dependent variable: Log of reservation wage		
Log of previous salary		0.237***
		[0.003]
Female	-0.113***	-0.073***
	[0.002]	[0.002]
Age 20-25	-0.02	-0.021
	[0.014]	[0.017]
Age 26-30	-0.114***	-0.086***
	[0.014]	[0.017]
Age 35 up	-0.135***	-0.108***
	[0.014]	[0.017]
On-the-job search	0.235***	0.138***
	[0.002]	[0.003]
Exp yrs: 1-4	0.118***	-0.052***
	[0.002]	[0.004]
Exp yrs: 5-10	0.398***	0.078***
	[0.003]	[0.005]
Exp yrs: 11-20	0.657***	0.237***
	[0.005]	[0.007]
Exp yrs: 20 above	0.817***	0.330***
	[0.018]	[0.016]
Freelance	-0.036**	-0.016
	[0.017]	[0.021]
Intern	-0.353***	-0.198***
	[0.039]	[0.055]
Part-time	-0.327***	-0.247***
	[0.009]	[0.013]
At education	0.081***	0.032
	[0.022]	[0.026]
Ed: M3 current	-0.369***	-0.289***
	[0.008]	[0.008]
Ed: M3	-0.413***	-0.337***
	[0.008]	[0.008]
Ed: HVC current	-0.321***	-0.253***
	[0.030]	[0.037]
Ed: HVC	-0.278***	-0.219***
	[0.006]	[0.006]
Ed: M6	-0.330***	-0.275***
	[0.007]	[0.007]
Ed: Bach current	-0.236***	-0.189***
	[0.023]	[0.028]
Ed: Master current	0.162***	0.141***
	[0.027]	[0.031]
Ed: Master	0.324***	0.217***
	[0.006]	[0.006]
Ed: PhD current	0.834***	0.730***
	[0.171]	[0.185]
Ed: Phd	0.16	0.322***
	[0.173]	[0.101]

GPA	0.027***	0.034***
	[0.002]	[0.002]
Technical vocational	-0.042***	-0.028***
	[0.006]	[0.006]
Commercial vocational	-0.061***	-0.046***
	[0.005]	[0.005]
Rajamakala bachelor	0.055***	0.051***
	[0.003]	[0.003]
Rajabhat bachelor	-0.093***	-0.068***
	[0.002]	[0.002]
General edu school	-0.118***	-0.083***
	[0.008]	[0.008]
Adult learning	-0.136***	-0.096***
	[0.010]	[0.010]
Know a European language	0.016	-0.002
	[0.025]	[0.038]
Know an Asian language	0.097***	0.094***
	[0.009]	[0.013]
Skill level in English	0.059***	0.038***
	[0.002]	[0.002]
Observations	152,429	108,089
Adjusted R-squared	0.485	0.603

Notes: ***1%, ** 5%, * 10% clustered standard errors. Other control variables are education qualification and province of work fixed effect. The data is taken from micro-data resume posting, accessed in July 2016. Omitted categories are those aged under 15, bachelor degree and no work experience. Column II also controls for occupation-type.

Table 4.1: Share of total job ads that target certain traits of applicants

	Percentage
Gender targeting	
Only female	14.47
Only male	17.93
Any gender	67.61
Gender missing	0
Age targeting	
16-19	9.68
20-14	38.39
25-29	19.5
30 or above	7.03
Any age	25.4
Total	41,312

Source: micro-data Job Portal No.2

-- End of Document --